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A Framework for Human-Aligned Sentiment Scoring in CBDC Text Analysis

A Graduation Project

Submitted by

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For the award of the degree

Of

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THESIS CERTIFICATE

This is to undertake that the Graduation Project titled **A FRAMEWORK FOR HUMAN-ALIGNED SENTIMENT SCORING IN CBDC TEXT ANALYSIS**, submitted by me to the School of Computing and Data Sciences, FLAME University, for the award of **BSc Data Science and Economics**, is a bona fide record of the research work done by me under the supervision of **Prof. Manoranjan Dash (Research Advisor) and Prof. Milind Sathye (Research Co-advisor)**. The contents of this Graduation Project, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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KEYWORDS CBDC; Central Bank Digital Currency; digital currency; e-CNY; financial sentiment; sentiment analysis; news sentiment; sentence-level sentiment; sentence-level annotation; dataset annotation; multi-annotator labeling; inter-annotator agreement; ICC; Krippendorff's Alpha; annotation quality; consensus scoring; human annotation benchmark; NLP; financial NLP; BERT; Sentence-BERT; Sentence Transformers; sentence embeddings; semantic similarity; cosine similarity; relevance filtering; text preprocessing; tokenization; NLTK; Bloomberg; Bloomberg news corpus; automated pipeline; document extraction; deduplication; supervised learning; regression model; model training; model evaluation; RMSE; R^2; country-wise sentiment trends; time-series sentiment; USA; Jamaica; Nigeria; CBDC discourse analysis

ABSTRACT

Central Bank Digital Currency (CBDC) represents a significant development in modern finance, with far-reaching implications for monetary policy, payment systems, and financial inclusion. Understanding how news media discusses CBDCs is therefore of considerable value to policymakers, financial institutions, and researchers seeking to monitor evolving public and institutional sentiment toward digital currency adoption. This project develops a sentence-level sentiment analysis framework for CBDC-related news, enabling sentiment to be measured in a granular, reproducible, and empirically rigorous manner.

The study is structured around two principal contributions. First, it constructs a manually annotated sentence-level dataset in which multiple annotators independently assign continuous sentiment scores to CBDC-related sentences. Annotation quality is systematically validated using established reliability metrics, including the Intraclass Correlation Coefficient (ICC) and Krippendorff's Alpha, ensuring that the resulting labels are consistent and suitable for model training. Second, it develops an automated

Bloomberg-based data pipeline that collects CBDC news articles, extracts and cleans raw text, tokenizes content at the sentence level using NLTK, and retains only semantically relevant sentences through embedding-based cosine similarity filtering.

The dataset encompasses CBDC-related coverage from three countries which are the United States, Jamaica, and Nigeria providing both geographic diversity and variation in CBDC implementation contexts. A Ridge Regression model is trained on the annotated dataset and evaluated against human annotation benchmarks. The inter-annotator agreement analysis demonstrates that sentiment labeling in financial policy text is a non-trivial task, characterized by inherent subjectivity that necessitates careful multi-annotator validation before model training.

Taken together, this work delivers three outputs: a validated CBDC sentiment dataset, a documented and reproducible annotation process, and a scalable Bloomberg-focused pipeline for large-scale sentiment analysis. These contributions provide a robust methodological foundation for future research on digital currency narratives, media framing of monetary policy, and the longitudinal dynamics of CBDC-related public discourse.

CONTENTS

		Page
List of Figures		**vii**
List of Tables		**viii**
1	**Introduction, Related Work, and Research Contributions**	**1**
1.1	Introduction	1
1.1.1	Central Bank Digital Currencies: Overview and Importance	1
1.1.2	Motivation for Sentiment Analysis	1
1.1.3	Problem Statement	2
1.1.4	Objectives	2
1.1.5	Financial and News Sentiment Analysis	3
1.1.6	Sentence-Level Versus Document-Level Analysis	3
1.1.7	Inter-Annotator Agreement in NLP	4
1.1.8	Transformer-Based Embeddings and Semantic Similarity	4
1.2	Related Work	5
1.2.1	Financial News Sentiment Analysis	5
1.2.2	CBDC Discourse and Sentiment in News Media	5
1.2.3	Sentence-Level Sentiment Annotation	6
1.2.4	Inter-Annotator Agreement in NLP	7
1.2.5	Semantic Filtering for Domain-Specific Corpora	7
1.2.6	Bloomberg as a Financial News Source	8
1.3	Research Contributions	8
1.3.1	Sentence-Level CBDC Dataset	8
1.3.2	Multi-Annotator Agreement Framework	8

2 Methodology 10

2.1 Dataset Construction and Preparation 10
2.2 Annotation Process and Reliability Validation 11
2.2.1 Consensus Score Computation 12
2.2.2 Dataset Composition 12
2.2.3 Inter-Annotator Agreement Analysis 12
2.3 Model Development and Evaluation 16
2.4 Semantic Filtering and Sentence Selection 17
2.4.1 Preprocessing 18
2.4.2 Embedding Generation 18
2.4.3 Query Construction 18
2.4.4 Similarity Computation and Filtering 18
2.4.5 Ranking and Top-K Selection 19
2.4.6 Context Window Inclusion 19
2.4.7 Semantic Deduplication of Articles 19
2.5 Sentiment Scoring Methodology 20
2.5.1 Sentence-Level Scoring 20
2.5.2 Document-Level Aggregation 20
2.5.3 Country-Level Aggregation 20
2.6 Data Collection for Large-Scale Analysis 21

3 Experimental Evaluation, Real-World Context, and Conclusion 22

3.1 Experimental Evaluation 22
3.1.1 Overview 22
3.1.2 Country-Wise Sentiment Trends (2020–2025) 22
3.1.3 Visualisation of Results 24
3.1.4 Summary of Findings 30

3.2 Real-World Reasons for Declining CBDC Sentiment 30
3.2.1 Low Adoption and Limited Usage 30
3.2.2 Privacy and Trust Concerns 30
3.2.3 Policy and Political Concerns 31
3.2.4 Technological and Infrastructure Challenges 31
3.2.5 Impact on the Banking System 31
3.2.6 Changing Media Narratives 31
3.3 Future Scope 32
3.4 Conclusion 33

Bibliography **35**

LIST OF FIGURES

Figure	Caption	Page
3.1	Multi-country sentiment trend comparison across USA, Nigeria, and Jamaica (2020–2025)	24
3.2	Year-by-year sentiment trend for Jamaica (2020–2025)	25
3.3	Year-by-year sentiment trend for Nigeria (2020–2025)	26
3.4	Dumbbell plot: 2020 vs. 2025 sentiment endpoints by country	27
3.5	Sentiment trends with uncertainty bands (2020–2025)	28
3.6	Country-year sentiment heatmap	29

LIST OF TABLES

Table	Caption	Page
3.1	Year-wise sentiment trend summary: USA, Nigeria, and Jamaica (2020–2025)	23

CHAPTER 1

INTRODUCTION, RELATED WORK, AND RESEARCH CONTRIBUTIONS

1.1 INTRODUCTION

1.1.1 Central Bank Digital Currencies: Overview and Importance

Central Bank Digital Currencies (CBDCs) are digital forms of money issued by central banks. Unlike cryptocurrencies, CBDCs are issued and controlled by national monetary authorities, making them a sovereign form of digital money. Many countries are currently testing or developing CBDCs with the aim of improving payment systems, reducing transaction costs, and supporting financial inclusion across diverse economic segments.

CBDCs are significant because they have the potential to fundamentally alter how individuals pay, save, and transfer money. Their implications extend beyond retail transactions, affecting monetary policy transmission, the structure of banking systems, and the efficiency of cross-border payments. Given the breadth of these effects, it is increasingly important to systematically study how CBDCs are discussed and framed in financial news media, as such coverage shapes public understanding and policy discourse.

1.1.2 Motivation for Sentiment Analysis

Financial news articles play an important role in influencing public understanding and shaping policy discussions. Shifts in media tone whether toward greater optimism or increased skepticism can signal changing confidence in an emerging financial technology and may precede or accompany broader adoption trends. Sentiment analysis provides a structured and reproducible method for measuring this tone at scale.

This project is motivated by the need to systematically track CBDC narratives over time using Bloomberg news text. By converting article sentences into quantitative sentiment

scores, it becomes possible to observe temporal and country-level trends that would not be readily apparent from manual reading alone. This approach transforms qualitative media content into a form amenable to rigorous empirical analysis.

1.1.3 Problem Statement

There are three practical challenges in CBDC news analysis that this project seeks to address. First, CBDC articles frequently contain mixed viewpoints within the same article, meaning that simple document-level sentiment labels can obscure important nuance and fail to capture the coexistence of supportive and critical perspectives. Second, sentiment in policy-focused financial text is inherently subjective, making high-quality label construction a non-trivial task that requires careful validation. Third, large-scale news collections are difficult to analyze manually, necessitating automated and scalable processing pipelines.

This project addresses these three problems through sentence-level scoring, which captures within-document sentiment variation; multi-annotator validation, which ensures label quality and reliability; and an automated Bloomberg-based processing workflow, which enables large-scale analysis without manual intervention.

1.1.4 Objectives

The primary objective of this project is to build a reliable and reproducible system for CBDC sentiment analysis using Bloomberg news documents. Specifically, the system is designed to:

1. Construct a sentence-level labeled dataset for CBDC-related news content, drawing on documents from Jamaica and Ghana as seed countries.

2. Measure inter-annotator agreement using multiple statistical metrics to validate the quality of the constructed labels.

3. Train a regression model to predict continuous sentiment scores from sentence-level text features.

4. Produce country-wise and year-wise sentiment trends from large-scale Bloomberg

data, enabling longitudinal and cross-national comparison.

1.1.5 Financial and News Sentiment Analysis

The field of financial sentiment analysis has evolved considerably over the past decade, progressing from lexicon-based methods to machine learning approaches and, more recently, to transformer-based models. Earlier methods relied on fixed word lists such as the Loughran-McDonald financial sentiment dictionary to assign polarity to documents. While interpretable, these approaches are limited in their ability to capture contextual meaning, domain-specific language, and subtle tonal shifts.

Newer transformer-based models, such as FinBERT and variants of BERT pre-trained on financial corpora, better capture the nuanced semantics of financial language. The majority of prior work in this space has focused on equity markets, earnings call transcripts, or general financial news. Comparatively fewer studies have targeted CBDC-specific narratives as a distinct domain. This project contributes to the literature by focusing specifically on CBDC discourse in Bloomberg coverage, a domain with its own terminology, policy context, and evolving sentiment landscape.

1.1.6 Sentence-Level Versus Document-Level Analysis

Document-level sentiment analysis assigns a single score to an entire article. While computationally simple, this approach can mask opposing views that coexist within the same text. For instance, a single Bloomberg article may simultaneously describe the macroeconomic benefits of a CBDC rollout in one paragraph while highlighting infrastructural risks or low adoption rates in another. A single document-level score would collapse these distinctions into a single value, losing potentially meaningful information.

Sentence-level analysis addresses this limitation by assigning individual scores to each sentence, thereby capturing finer-grained sentiment variation within documents. This granularity also enables more precise aggregation across time periods and countries.

For CBDC policy news where articles are typically nuanced, multi-perspective, and technically detailed sentence-level analysis is the more appropriate analytical unit.

1.1.7 Inter-Annotator Agreement in NLP

Sentiment labeling is an inherently subjective task, and disagreement between annotators can introduce noise into training data, reducing the reliability of downstream models. Inter-annotator agreement analysis provides a principled way to evaluate the consistency of labels before they are used for model training. High agreement indicates that the labeling scheme is well-defined and that annotators are applying it consistently; low agreement suggests ambiguity in the annotation guidelines or the nature of the text itself.

This project employs a multi-annotator setup in which three human annotators and a Large Language Model (LLM) independently label each sentence. Agreement is then quantified using multiple complementary metrics that include Mean Absolute Error (MAE), Bias, Pearson correlation, Krippendorff's Alpha, and the Intraclass Correlation Coefficient (ICC) before any model training takes place. This validation step strengthens confidence in the sentiment scores used for downstream trend analysis.

1.1.8 Transformer-Based Embeddings and Semantic Similarity

Transformer-based language models represent sentences as dense, high-dimensional vectors that encode contextual meaning rather than simple keyword frequency. These representations allow downstream models to distinguish between sentences that use similar vocabulary in different ways, which is particularly valuable in financial language where subtle differences in wording can carry significant meaning.

In this project, transformer embeddings serve two roles. First, they are used to compute semantic similarity between candidate sentences and a CBDC-specific query, enabling the filtering and selection of the most relevant sentences from large document collections. Second, embedding-based features support sentence-level sentiment prediction on Bloomberg documents, providing the model with richer representations of sentence

meaning than traditional bag-of-words or TF-IDF features would allow. Together, these capabilities make transformer embeddings a central component of the proposed analysis pipeline.

1.2 RELATED WORK

1.2.1 Financial News Sentiment Analysis

Sentiment analysis of financial text has been an active area of NLP research, motivated by the value of extracting market perceptions, policy signals, and investor expectations from news corpora. Early approaches relied on domain-specific lexicons (Loughran & McDonald, 2011), while later work moved toward supervised machine learning and, eventually, pre-trained transformer models. A landmark contribution in this space is FinBERT (Araci, 2019), a variant of BERT fine-tuned on financial corpora that significantly improved the state of the art on sentiment classification benchmarks such as Financial PhraseBank (Malo et al., 2014). Subsequent work by Olaniyan et al. (2021) demonstrated that even small manually labelled samples of financial news, when combined with optimised BERT-based architectures, can yield high-quality sentiment extraction without requiring large computational resources. More recently, Shen and Zhang (2024) compared large language models (GPT-4o) and FinBERT on news articles and financial reports, finding that few-shot prompting with GPT-4o can match a well fine-tuned FinBERT on specialised tasks. These advances motivate the use of transformer-based models in our project for automated sentiment scoring of CBDC-related sentences.

1.2.2 CBDC Discourse and Sentiment in News Media

Research on CBDCs has grown substantially, examining policy adoption, public trust, privacy concerns, and financial inclusion (Auer et al., 2021). In the news media domain, Hofmann et al. (2025) developed jurisdiction-level CBDC sentiment indices from both central bank communications and media coverage using NLP, finding that CBDC sentiment in emerging market economies diverged noticeably from that of advanced

economies—a pattern directly relevant to our cross-country comparative design. Alves (2025) employed ChatGPT 4.0 to score CBDC-related news sentiment from 2018 to 2024, constructing a CBDCSX index that tracked how political and launch-related events shifted overall sentiment polarity. At the social media level, a comparative study of UK CBDC discourse on X (Anonymous, 2025) evaluated DistilBERT, RoBERTa, and XLM-RoBERTa for sentiment classification of tweets, finding that RoBERTa trained for three epochs best generalised to unseen data. Conlon et al. (2024) examined how sentiment towards CBDCs, measured through social media, responded to central bank announcements, geopolitical events, and cryptocurrency market shocks. Our project differs from these works by operating at the sentence level on Bloomberg news articles rather than document-level social media data, and by incorporating rigorous multi-annotator validation before any model is applied.

1.2.3 Sentence-Level Sentiment Annotation

Sentence-level analysis is preferred over document-level scoring when an article contains mixed viewpoints, since aggregating polarity at the document level loses fine-grained stance information. The Financial PhraseBank (Malo et al., 2014) established the standard approach for financial sentence-level annotation: approximately 5,000 sentences drawn from financial news were independently labelled by up to 16 annotators with finance backgrounds, and the dataset is partitioned by annotator agreement thresholds (≥50%, ≥66%, ≥75%, 100%). This multi-threshold structure demonstrates how agreement level directly affects dataset quality and downstream model performance. Zafar (2025) extended sentence-level labelling to CBDC speech data, applying a domain-adapted BERT architecture to classify 5,376 CBDC sentences from BIS speeches by type, stance, sentiment, and discourse category. Our work follows this sentence-level paradigm but applies it to Bloomberg news rather than central bank speeches, and uses continuous sentiment scoring rather than categorical labels.

1.2.4 Inter-Annotator Agreement in NLP

Since sentiment in financial and policy text is inherently subjective, the reliability of human annotations must be quantitatively validated before labels are used for model training. Artstein (2017) provides a comprehensive survey of inter-annotator agreement coefficients, covering the mathematical assumptions behind Krippendorff's α, Scott's π, and Cohen's κ, and argues that weighted α-like coefficients are more appropriate than κ-like measures for many corpus annotation tasks. Hayes and Krippendorff (2007) advocated for Krippendorff's α as a universal reliability standard because it accommodates any number of annotators, any measurement level (nominal, ordinal, interval), and missing data. Lawrence et al. (2025) recently introduced AmbiSent, a sentiment dataset designed to study interpretive complexity, and found that inter-rater agreement (measured by Krippendorff's α) was significantly affected by the degree of instruction detail provided to annotators—highlighting why careful annotation guidelines are essential. Our project employs three complementary agreement metrics (MAE, ICC, and Krippendorff's α) across four annotators to provide a multi-faceted assessment of label quality, extending standard single-metric reporting in prior work.

1.2.5 Semantic Filtering for Domain-Specific Corpora

Financial news corpora assembled via keyword search frequently contain articles that mention terms like "digital currency" or "digital yuan" without substantively discussing CBDCs. Addressing this noise is critical for dataset precision. Reimers and Gurevych (2019) introduced Sentence-BERT (SBERT), a modification of BERT using siamese networks to produce semantically meaningful sentence embeddings directly comparable via cosine similarity. SBERT made large-scale semantic filtering computationally tractable by reducing pairwise comparison time from hours to seconds. Concurrently, Devlin et al. (2019) established the general BERT architecture upon which both domain-specific models and sentence embedding approaches are built. In the CBDC context, Hofmann et al. (2025) applied keyword filtering combined with sentence-level subject analysis to discard irrelevant articles from their media corpus. Our pipeline employs

cosine similarity between sentence embeddings and a CBDC seed document to filter out tangentially related articles before annotation, directly operationalising the approach supported by Reimers and Gurevych (2019).

1.2.6 Bloomberg as a Financial News Source

Studies that build financial sentiment corpora from curated editorial sources rather than noisy social media or API-scraped feeds benefit from higher consistency in writing style, terminology, and factual accuracy. Lefort et al. (2025) constructed FinMarBa, a large-scale financial sentiment dataset drawn from Bloomberg Market Wraps, and noted that Bloomberg provides reliable high-quality news closely followed by market participants, with coverage spanning global economic events—an advantage over region-specific or social media corpora. Our project similarly restricts its corpus to Bloomberg articles. While this improves source consistency and domain specificity, it limits heterogeneity and generalisability; we acknowledge this as a deliberate trade-off that makes the dataset suitable for controlled analysis of CBDC coverage within a professional financial news context.

1.3 RESEARCH CONTRIBUTIONS

This project makes the following key contributions to the field:

1.3.1 Sentence-Level CBDC Dataset

We introduce a manually annotated dataset of 500 CBDC-related sentences, scored by four independent annotators, establishing a novel resource for CBDC sentiment research. This dataset provides a foundation for future work in digital currency sentiment analysis, supported by rigorous annotation guidelines and quality measurements.

1.3.2 Multi-Annotator Agreement Framework

We implement and evaluate three inter-annotator agreement metrics—Mean Absolute Error (MAE), Intraclass Correlation Coefficient (ICC), and Krippendorff's Alpha—to

rigorously assess annotation quality and annotator bias. This framework offers transparent evaluation of consensus across multiple expert and automated annotators. By introducing a multi-annotator scoring setup with agreement analysis, the collected labels are not merely aggregated but validated for consistency.

Additional contributions of this framework include:

1. **Cross-country CBDC trend analysis**, enabling comparison of how sentiment evolves across different national contexts and over time.

2. **Transparent and reproducible workflow documentation**, making the dataset and methodology more accessible and replicable for future research.

3. **Semantic similarity filtering** to remove irrelevant articles, improving retrieval precision within a keyword-heavy search space.

CHAPTER 2

METHODOLOGY

2.1 DATASET CONSTRUCTION AND PREPARATION

The study began with the creation of a controlled, high-quality seed dataset comprising 150 sentences related to Central Bank Digital Currencies (CBDCs). These sentences were sourced exclusively from official and credible publications, including central bank reports, policy documents, and authoritative financial articles. To ensure consistency and reduce heterogeneity in the initial phase, the dataset was restricted to two countries: **Jamaica** and **Ghana**—both of which have been early movers in CBDC deployment (the JAM-DEX and the e-Cedi, respectively), making them ideal case studies for capturing authentic policy sentiment.

Sentence	You	LLM	Prof	Dash	FinalScore
Ghana is currently in the pilot phase of its central bank digital currency, known as the e-Cedi.	0.5	0.5	0.6	0.55	0.5375
The Bank of Ghana launched the e-Cedi pilot to test the feasibility of a retail digital currency	0.5	0.5	0.7	0.55	0.5625
The central bank views the e-Cedi as a tool to promote financial inclusion and digital payments.	0.75	0.6	0.8	0.65	0.7
Stakeholders in the financial sector have observed meaningful strides from the central bank regarding the introduction of the eCedi.	0.75	0.6	0.7	0.85	0.725
The introduction and full implementation of the eCedi, together with broader digital payment reforms, will be crucial in strengthening Ghana's monetary sovereignty.	0.65	0.75	0.6	0.7	0.675
The First Deputy Governor of the Bank of Ghana described eCedi as a key pillar of Ghana's digital future.	0.7	0.75	0.8	0.85	0.775
The central bank digital currency will create a more efficient payment ecosystem that proves faster, safer, and more reliable for individuals and businesses.	0.8	0.75	0.9	0.8	0.8125
Early and deliberate investments in digital systems have created an enabling environment for successfully adopting a CBDC at scale.	0.85	0.6	0.6	0.8	0.7125
Robust security measures and minimal transaction costs will be critical for successful adoption of the eCedi.	0.5	0.6	0.5	0.6	0.55
The pilot revealed significant implementation challenges as it moves closer to rolling out the eCedi.	0.32	0.3	0.3	0.25	0.2925
Stakeholder feedback suggests that public trust and fee considerations could significantly influence adoption rates.	0.5	0.6	0.3	0.5	0.475
Despite the popularity of mobile money, the top bank has been pushing the eCedi central bank digital currency, whose pilot has been ongoing this year.	0.5	0.5	0.5	0.75	0.5625

Each sentence was manually extracted and curated to ensure that it represented a complete and meaningful unit of analysis. Care was taken to include sentences that directly reflected opinions, observations, or factual statements related to CBDC adoption, infrastructure, policy, risks, and economic implications. The objective at this stage was to construct a

clean and reliable foundational dataset that could be used for both annotation and model training.

Following the initial phase, the dataset was expanded to **500 sentences**, maintaining the same country focus. This expansion significantly improved the diversity of the dataset by incorporating a wider range of linguistic structures, tones, and contextual nuances. The expanded dataset included both supportive narratives (e.g., benefits of financial inclusion and payment efficiency) and critical perspectives (e.g., low adoption rates and infrastructural challenges), thereby ensuring a balanced representation of sentiment across the full spectrum of discourse.

2.2 ANNOTATION PROCESS AND RELIABILITY VALIDATION

The expanded dataset was subjected to a rigorous multi-annotator labeling process involving four independent annotators: three human annotators (Annotator 1, Annotator 2, and Annotator 3) and one automated annotator in the form of a Large Language Model (LLM). The inclusion of the LLM as a fourth annotator was intended to complement human judgment and introduce a computationally consistent perspective into the labeling process.

Each annotator evaluated every sentence independently and assigned a *sentiment score* on a continuous scale from 0 to 1, where:

- **0** indicates strongly negative sentiment (e.g., a sentence expressing severe risk, failure, or opposition to CBDCs)
- **0.5** indicates neutral sentiment (e.g., a purely factual or descriptive statement with no clear evaluative stance)
- **1** indicates strongly positive sentiment (e.g., a sentence praising adoption, financial inclusion, or policy success)

The use of a continuous scale, rather than discrete categories, allows for finer-grained sentiment distinctions and supports regression-based modeling in downstream tasks.

2.2.1 Consensus Score Computation

The final consensus score for each sentence was computed as the *arithmetic mean* of the four annotators' scores, ensuring equal weighting across all annotators:

$$\text{FinalScore} = \frac{\text{Score}_{A1} + \text{Score}_{A2} + \text{Score}_{A3} + \text{Score}_{\text{LLM}}}{4} \tag{2.1}$$

where $A1$, $A2$, and $A3$ denote the three human annotators and LLM denotes the automated annotator. This averaging approach is widely used in multi-annotator frameworks. By pooling scores across human experts and an automated annotator, the mean helps *reduce idiosyncratic noise* introduced by any single rater and produces a more stable, representative label for each sentence. It also implicitly assumes that individual biases are random and partially cancel out when averaged.

2.2.2 Dataset Composition

Following preprocessing and filtering, the final working dataset consisted of **500 high-quality sentences**, each structured with the following attributes:

- **Sentence**: The CBDC-related text excerpt
- **A1, A2, A3, LLM**: Individual annotator scores on the [0, 1] scale, corresponding to the three human annotators and the LLM respectively
- **FinalScore**: The arithmetic mean of all four annotator scores, used as the ground-truth label
- **Ref A1, Ref A2, Ref A3, Ref LLM**: Reference scores computed as the mean of the *other three* annotators, used for agreement analysis

This structured format enabled both supervised model training and detailed statistical evaluation of annotator behavior.

2.2.3 Inter-Annotator Agreement Analysis

Ensuring that annotators agree with one another is critical before any labeled dataset can be considered reliable. If annotators interpret sentences very differently, the resulting

labels would be noisy and unsuitable for training a robust model. To validate the quality of the annotation, multiple complementary agreement metrics were computed, each capturing a different dimension of reliability.

Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) measures how far an individual annotator's scores deviate, on average, from the consensus of the remaining three annotators. It serves as an indicator of *annotator consistency*: a lower MAE means the annotator is more closely aligned with the group consensus.

For each annotator X, a reference score Ref_X is first computed as the mean of scores assigned by the other three annotators. The MAE is then defined as:

$$\text{MAE}_X = \frac{1}{N} \sum_{i=1}^{N} |X_i - (\text{Ref}_X)_i| \tag{2.2}$$

where N is the total number of sentences. **Results:**

- LLM MAE $\approx$ 0.087 — the lowest among all annotators, indicating the highest consistency with the group consensus
- Annotator 1 MAE $\approx$ 0.091 — very close to the LLM, indicating strong alignment
- Annotator 3 MAE $\approx$ 0.094 — slightly higher, but still within a narrow band
- Annotator 2 MAE $\approx$ 0.106 — the highest, suggesting slightly greater variability in interpretation relative to the group

Overall, all MAE values are small (well below 0.15), indicating that no annotator deviates substantially from the group consensus. In practical terms, a MAE of 0.1 on a 0–1 scale means an annotator's scores differ from the reference by only 0.1 on average—a level of disagreement considered acceptable in sentiment annotation tasks.

Bias Analysis

While MAE measures the *magnitude* of deviation, it does not reveal whether annotators systematically score higher or lower than their peers. Bias analysis addresses this by computing the *signed mean deviation*:

$$\text{Bias}_X = \frac{1}{N} \sum_{i=1}^{N} (X_i - (\text{Ref}_X)_i) \tag{2.3}$$

A positive bias indicates a tendency to score sentiment higher than peers; a negative bias indicates the opposite. **Results:**

- LLM Bias ≈ -0.001
- Annotator 1 Bias ≈ -0.004
- Annotator 2 Bias ≈ -0.008
- Annotator 3 Bias ≈ -0.007

All four annotators exhibited bias values extremely close to zero, indicating no systematic tendency to over- or under-score sentiment. The slight negative values suggest a marginally conservative tendency across all annotators, but this is negligible in magnitude. This result increases confidence that the final consensus scores are free from directional distortion.

Pearson Correlation Matrix

The Pearson correlation coefficient measures the *linear association* between two annotators' scores across all sentences. A value of +1 indicates perfect agreement, 0 indicates no linear relationship, and −1 indicates perfect disagreement. Pairwise correlations were computed for all annotator pairs:

- A1–A3 ≈ 0.75 — the highest pairwise correlation, suggesting strong agreement between these two human annotators
- A1–A2 ≈ 0.74

- LLM–A1 ≈ 0.72
- LLM–A3 ≈ 0.71
- A2–A3 ≈ 0.69
- LLM–A2 ≈ 0.68 — the lowest pairwise correlation, though still indicating moderate agreement

Values in the range of 0.68–0.75 reflect **moderate to high agreement**, which is consistent with expectations for a nuanced, domain-specific sentiment task where some interpretive subjectivity is inevitable. Notably, the LLM correlates comparably with all human annotators, confirming its viability as a reliable automated labeler in this context.

Krippendorff's Alpha

Krippendorff's Alpha (α) is a general-purpose reliability measure that accounts for chance agreement, scales of measurement, and missing data. It is particularly suited to ordinal and interval-level annotations, making it appropriate here given the continuous [0, 1] scale. The metric is defined as:

$$\alpha = 1 - \frac{D_o}{D_e} \tag{2.4}$$

where D_o is the observed disagreement among annotators and D_e is the disagreement expected by chance. An α of 1 indicates perfect reliability; $\alpha = 0$ indicates agreement at chance level; negative values suggest systematic disagreement.

Result: $\alpha \approx 0.697$

An α of 0.697 falls within the range conventionally described as **moderate to acceptable reliability**. Krippendorff (2004) suggests a threshold of $\alpha \geq 0.667$ for tentative conclusions and $\alpha \geq 0.800$ for high-stakes decisions. Given the inherent subjectivity of sentiment annotation in specialized financial domains, an α of approximately 0.70 is a satisfactory and commonly reported level in NLP annotation studies.

Intraclass Correlation Coefficient (ICC)

The Intraclass Correlation Coefficient (ICC) is a widely used reliability metric in psychometrics that assesses both the *consistency* and *absolute agreement* among raters. Unlike Pearson correlation, ICC is sensitive to systematic differences in scale use across annotators (i.e., one annotator always scoring 0.1 higher than another would reduce ICC even if the rank ordering is identical). The ICC(3,k) form was used, which is appropriate for a fixed set of raters and focuses on the reliability of the mean of all raters' judgments.

Result: ICC ≈ 0.78

An ICC of 0.78 falls in the range of **good to excellent agreement** according to established benchmarks (Koo & Mae, 2016: poor < 0.50; moderate 0.50–0.75; good 0.75–0.90; excellent > 0.90). This value confirms that annotators not only agree on the *direction* of sentiment but also on the *magnitude* of their scores, providing strong evidence of annotation reliability.

Taken together, the suite of agreement metrics—MAE, Bias, Pearson r, Krippendorff's α, and ICC—converge on the conclusion that the annotation process was statistically reliable, consistent, and free from systematic bias, making the dataset suitable for model training.

2.3 MODEL DEVELOPMENT AND EVALUATION

The annotated dataset was divided into training and testing subsets using an **80:20 split**, with 400 sentences for training and 100 for evaluation, ensuring that model performance was assessed on entirely unseen data.

A variety of regression models were explored for predicting continuous sentiment scores from text features, including linear regression, Lasso, ElasticNet, and tree-based methods. Among these, **Ridge Regression** was identified as the best-performing model. Ridge

Regression is a form of penalized least-squares that adds an ℓ_2 regularization term to the standard regression objective:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \tag{2.5}$$

where $\lambda > 0$ is the regularization strength. The choice of Ridge Regression was motivated by its ability to:

- **Handle multicollinearity effectively** — text-based features (e.g., TF-IDF vectors) are often highly correlated, and Ridge shrinks correlated feature coefficients together rather than arbitrarily selecting one
- **Prevent overfitting through regularization** — the ℓ_2 penalty discourages large coefficients, improving generalization to unseen sentences
- **Perform reliably on relatively small datasets** — unlike deep learning models, Ridge Regression provides stable estimates even with limited training data

Model performance was evaluated using two complementary metrics:

- **Root Mean Squared Error (RMSE): 0.087** — RMSE measures the average magnitude of prediction error in the same units as the target variable. An RMSE of 0.087 on a [0, 1] scale indicates that predicted sentiment scores deviate from the true scores by approximately 0.087 on average—a strong result indicating high predictive accuracy.
- R^2 **(Coefficient of Determination): 0.602** — R^2 represents the proportion of variance in the sentiment scores explained by the model. An R^2 of 0.602 means the model accounts for approximately **60% of the variability** in sentiment scores, which is a reasonable fit given the inherent subjectivity of sentiment in specialized financial text and the relatively modest size of the training set.

Based on these metrics, Ridge Regression was selected as the final model for large-scale sentiment scoring.

2.4 SEMANTIC FILTERING AND SENTENCE SELECTION

To ensure that only genuinely CBDC-relevant sentences were passed to the sentiment scoring pipeline, a multi-stage semantic filtering process was applied.

2.4.1 Preprocessing

Before semantic analysis, sentences were filtered based on length. Sentences containing **fewer than six words** were discarded, as they are typically too short to carry meaningful sentiment-laden content (e.g., headers, fragment clauses, or standalone numerical values).

2.4.2 Embedding Generation

Sentence-level dense vector representations were generated using the pre-trained transformer model `all-MiniLM-L6-v2` from the Sentence Transformers library. This model produces 384-dimensional embeddings optimized for semantic similarity tasks. Its lightweight architecture makes it computationally efficient while maintaining strong performance on sentence-level similarity benchmarks.

2.4.3 Query Construction

A domain-specific CBDC query text was manually constructed to serve as the semantic anchor for filtering. It incorporates a broad vocabulary of CBDC-related concepts to maximize recall:

```
central bank digital currency cbdc digital rupee digital yuan enaira jamdex
digital payments modernization cashless economy mobile wallet adoption
financial inclusion instant payments reduced transaction costs
retail payments government transfers merchant acceptance
secure transparent traceable digital money infrastructure
```

This composite query ensures that sentences addressing diverse facets of the CBDC ecosystem—technological, economic, and policy-related—are captured.

2.4.4 Similarity Computation and Filtering

Cosine similarity was computed between each sentence embedding and the query embedding. Cosine similarity measures the angle between two vectors in high-dimensional space, with values ranging from -1 (opposite directions) to $+1$

(identical direction), and is particularly well-suited for comparing sentence embeddings. Sentences whose cosine similarity with the query exceeded a predefined threshold were retained for further analysis; those below the threshold were discarded as semantically irrelevant.

2.4.5 Ranking and Top-K Selection

Among the filtered sentences, the top $K = 80$ sentences were selected based on their cosine similarity scores. This step ensures that only the most semantically relevant content is forwarded to the sentiment model, reducing noise from marginally related sentences.

2.4.6 Context Window Inclusion

To preserve local discourse structure and improve interpretability, a *context window* of size 1 was applied around each selected sentence, including the immediately preceding and succeeding sentences. This ensures that selected sentences are not interpreted in isolation, but within their immediate narrative context—which can be important for ambiguous or elliptical statements.

2.4.7 Semantic Deduplication of Articles

To prevent redundancy in the final corpus, a semantic deduplication step was implemented at the *document level* using BERT-based title embeddings:

- Article titles were converted into dense vector embeddings
- Pairwise cosine similarity was computed across all article title pairs
- Articles with a pairwise cosine similarity exceeding a threshold of **0.9** were considered near-duplicates; one copy was discarded

A threshold of 0.9 is deliberately stringent, ensuring that only articles with nearly identical titles—and thus highly overlapping content—are removed, while preserving thematically related but distinct articles. Logging was used throughout to track the total number of articles processed, removed, and retained, providing transparency and reproducibility.

2.5 SENTIMENT SCORING METHODOLOGY

2.5.1 Sentence-Level Scoring

Each sentence passing the semantic filter was scored independently using the trained Ridge Regression model. The model takes a sentence's text features as input and produces a *continuous sentiment score* in the range [0, 1], where values near 0 indicate negative sentiment, values near 0.5 indicate neutrality, and values near 1 indicate positive sentiment.

2.5.2 Document-Level Aggregation

Document-level sentiment was computed as the *unweighted arithmetic mean* of all sentence-level scores within a given document. This approach assumes that each sentence contributes equally to the overall document sentiment—a reasonable assumption in the absence of article structure metadata (e.g., headline vs. body text).

$$\text{Sentiment}_{\text{doc}} = \frac{1}{|S|} \sum_{s \in S} \text{Score}(s) \tag{2.6}$$

where S is the set of sentences in the document and $\text{Score}(s)$ is the predicted sentiment score for sentence s.

2.5.3 Country-Level Aggregation

Country-level sentiment was computed as the *unweighted arithmetic mean* of all document-level scores associated with a given country. This aggregation produces a single scalar representing the overall sentiment landscape for each country's CBDC discourse during the study period.

$$\text{Sentiment}_{\text{country}} = \frac{1}{|D|} \sum_{d \in D} \text{Sentiment}_{\text{doc}}(d) \tag{2.7}$$

where D is the set of documents associated with a given country.

2.6 DATA COLLECTION FOR LARGE-SCALE ANALYSIS

For the final large-scale analysis, documents were collected exclusively from **Bloomberg**, a globally recognized financial news platform with comprehensive coverage of monetary policy, central banking, and digital currency developments. Restricting the source to a single, authoritative outlet ensures consistency in journalistic style, publication standards, and topical focus across all collected documents.

A **country-specific keyword strategy** was employed during data collection to ensure relevance and representativeness of the dataset. For each country under study, a tailored set of search terms was constructed to capture the distinct terminology used in that country's CBDC discourse (e.g., referencing country-specific CBDC names, central bank identifiers, and locally relevant financial concepts). This strategy maximizes the likelihood that collected documents reflect genuine national-level sentiment rather than generic global CBDC commentary.

CHAPTER 3

EXPERIMENTAL EVALUATION, REAL-WORLD CONTEXT, AND CONCLUSION

3.1 EXPERIMENTAL EVALUATION

3.1.1 Overview

This section presents the results of the sentiment analysis conducted across CBDC-related news articles spanning the period 2020 to 2025. The evaluation is structured around country-level sentiment trends, with the objective of identifying whether public and media discourse surrounding CBDCs has shifted over time. Sentiment scores are continuous values in the range $[0, 1]$, where lower values indicate more negative sentiment and higher values indicate more positive sentiment. Results are reported at the annual level, aggregated from document- and sentence-level predictions generated by the trained Ridge Regression model.

3.1.2 Country-Wise Sentiment Trends (2020–2025)

Countries with Full Year-Wise Series

Three countries—USA, Nigeria, and Jamaica—had sufficient data coverage to support a complete year-by-year trend analysis across the full 2020–2025 study window. Table 3.1 summarises the 2020 baseline mean, the 2025 endpoint mean, the absolute change (Delta), and the percentage change for each country.

Table 3.1: Year-wise sentiment trend summary: USA, Nigeria, and Jamaica (2020–2025)

Country	**2020 Mean**	**2025 Mean**	**Delta**	**Percent Change**
USA	0.410	0.292	−0.118	−28.8%
Nigeria	0.380	0.278	−0.102	−26.9%
Jamaica	0.384	0.359	−0.025	−6.6%

All three countries record lower mean sentiment scores in 2025 compared to their 2020 baselines, indicating a consistent downward shift in CBDC-related media sentiment over the study period. The USA exhibits the steepest decline, with a reduction of 0.118 points (−28.8%), followed closely by Nigeria at −0.102 points (−26.9%). Jamaica shows a comparatively modest decline of 0.025 points (−6.6%), suggesting that sentiment in its CBDC discourse has remained more stable, though still trending negatively overall.

3.1.3 Visualisation of Results

The following figures provide visual support for the quantitative findings reported above. Each figure is discussed in the context of what it reveals about country-level sentiment dynamics over the study period.

Figure 1: Multi-Country Sentiment Trend Comparison

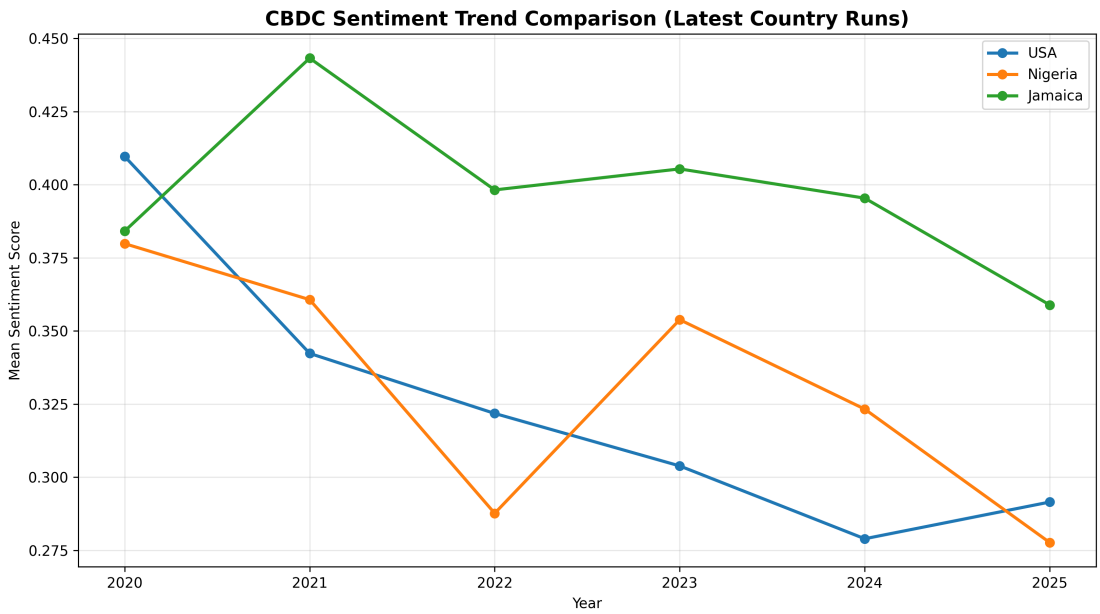

Figure 3.1: Multi-country sentiment trend comparison across USA, Nigeria, and Jamaica (2020–2025)

Figure 3.1 presents the primary multi-country trend comparison. All three lines—representing USA, Nigeria, and Jamaica—close below their respective 2020 starting points by 2025, visually confirming the declining pattern reported in Table 3.1. The steeper gradients of the USA and Nigeria lines relative to Jamaica are consistent with the larger absolute deltas observed for those countries.

Figure 2: Jamaica Year-by-Year Sentiment Trend

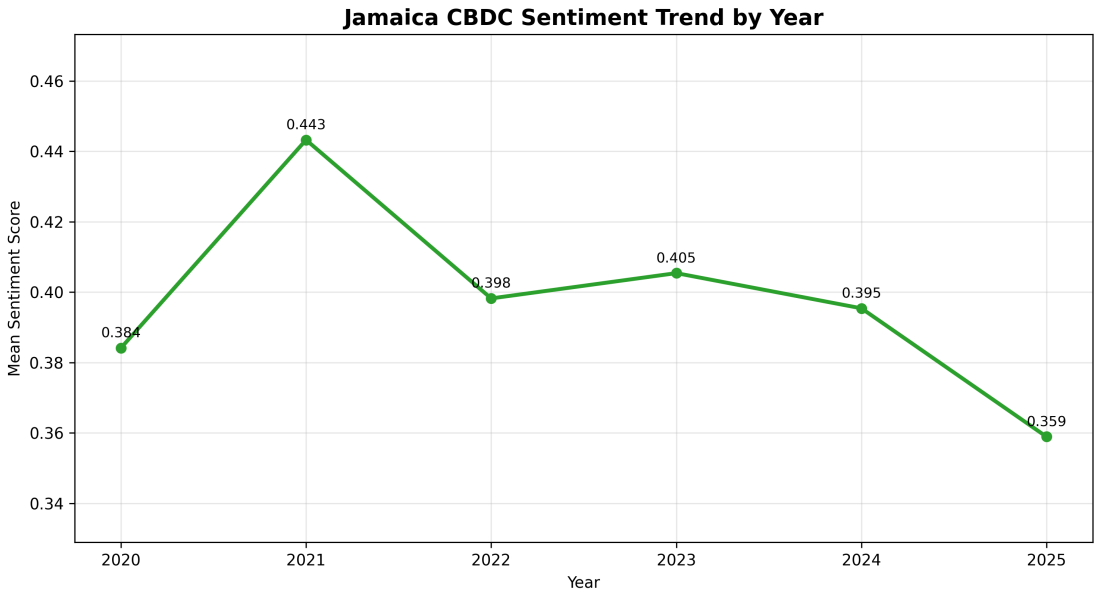

Figure 3.2: Year-by-year sentiment trend for Jamaica (2020–2025)

Figure 3.2 shows that Jamaica's sentiment peaks in 2021, coinciding with the initial launch and promotion of the JAM-DEX, before gradually declining through the remainder of the study period. The relatively contained decline of −6.6% distinguishes Jamaica from the sharper downturns seen in larger economies, possibly reflecting the smaller volume of critical financial press coverage directed at a smaller-scale CBDC programme.

Figure 3: Nigeria Year-by-Year Sentiment Trend

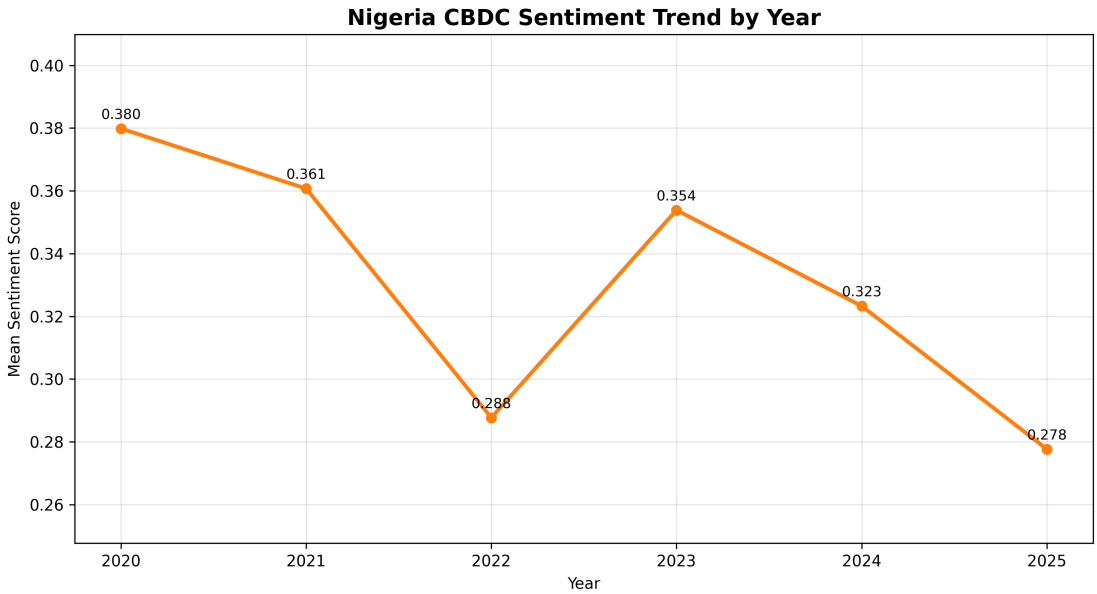

Figure 3.3: Year-by-year sentiment trend for Nigeria (2020–2025)

Figure 3.3 reveals a more volatile trajectory for Nigeria. Sentiment drops steeply to a trough around 2022—likely reflecting early challenges surrounding the eNaira's rollout and low adoption rates—before recovering partially, then declining again through to 2025. The overall net decline of −26.9% is comparable in magnitude to the USA, suggesting persistent negative sentiment despite intermittent recoveries.

Figure 4: 2020 vs. 2025 Dumbbell Comparison

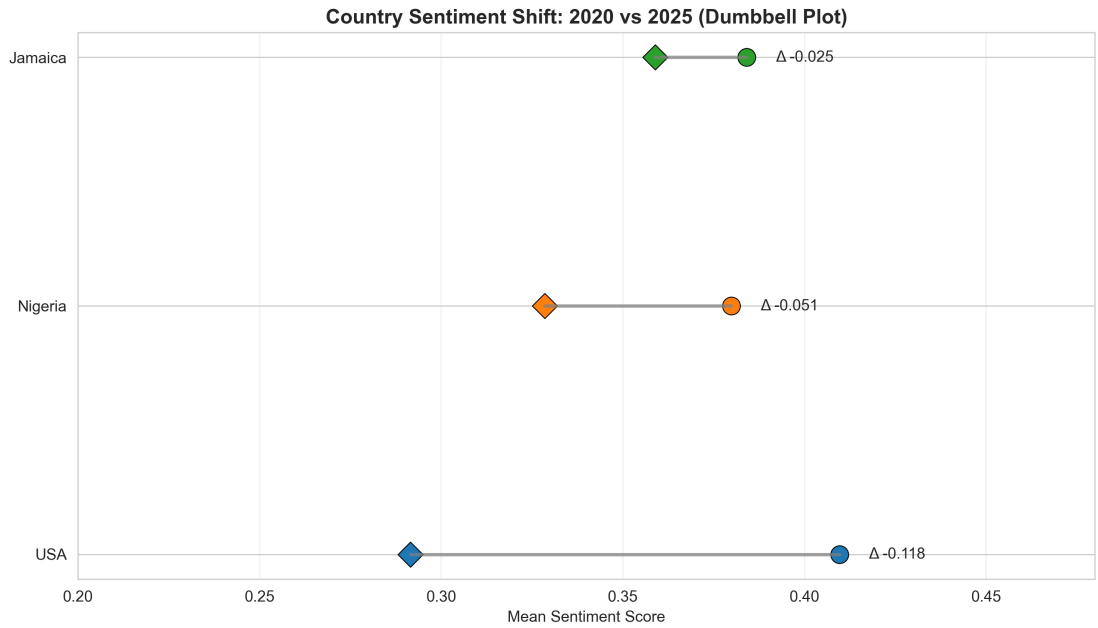

Figure 3.4: Dumbbell plot: 2020 vs. 2025 sentiment endpoints by country

Figure 3.4 offers the most direct start-to-end comparison. Each dumbbell segment connects a country's 2020 mean to its 2025 mean, with the direction and length of the segment representing the sign and magnitude of change. All segments point in the negative direction, reinforcing the conclusion that sentiment has declined across all three fully trended countries over the five-year window.

Figure 5: Sentiment Trends with Uncertainty Bands

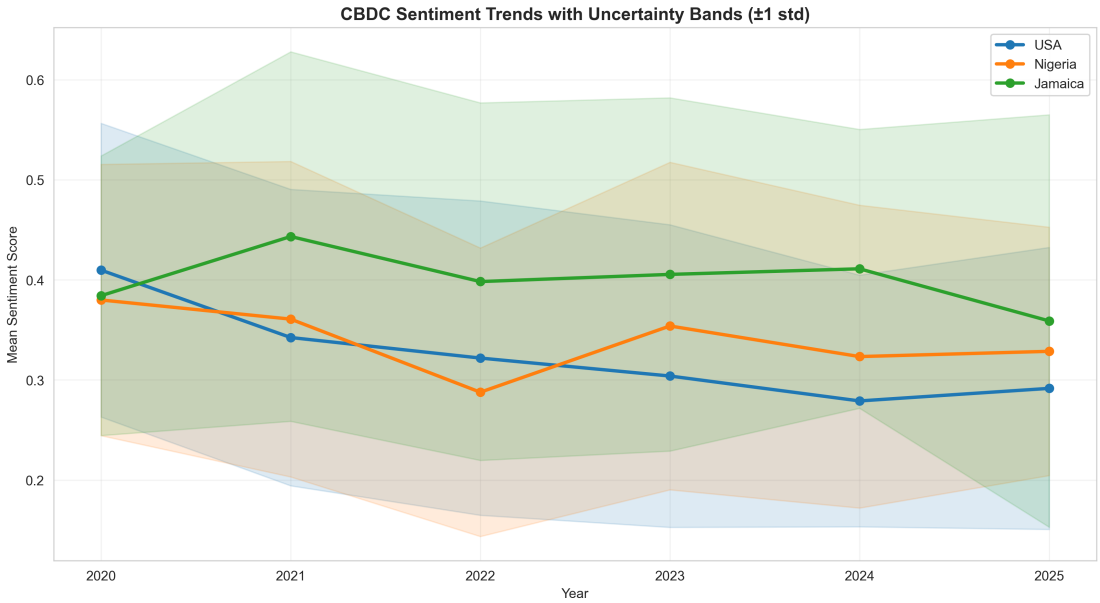

Figure 3.5: Sentiment trends with uncertainty bands (2020–2025)

Figure 3.5 augments the trend lines with uncertainty envelopes derived from the variance in annual document-level scores. The downward trends remain clearly visible even accounting for uncertainty, indicating that the observed declines are not artefacts of a small number of outlier documents. The non-trivial width of the bands, however, reflects meaningful year-to-year heterogeneity in article-level sentiment, underscoring the importance of aggregating over sufficiently large document samples.

Figure 6: Country-Year Sentiment Heatmap

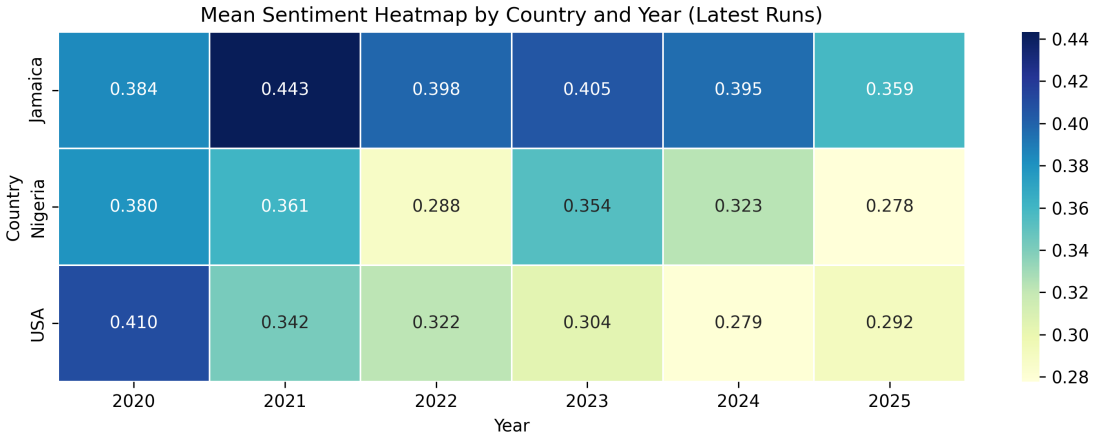

Figure 3.6: Country-year sentiment heatmap

Figure 3.6 presents sentiment scores in a compact matrix format, with countries on one axis and years on the other. Darker cells—corresponding to lower sentiment values—are concentrated in the later years (2024–2025) for USA and Nigeria, providing an immediately interpretable visual summary of where and when sentiment deteriorated most sharply.

3.1.4 Summary of Findings

The experimental evaluation yields the following key findings:

- **Universal decline across fully trended countries.** USA, Nigeria, and Jamaica all exhibit lower mean CBDC sentiment in 2025 than in 2020, indicating a broad deterioration in media and public discourse over the five-year study period.

- **Magnitude of decline varies by country.** The USA (−28.8%) and Nigeria (−26.9%) experience sharp declines, while Jamaica's decline is more moderate (−6.6%), potentially reflecting differences in CBDC scale, adoption challenges, and press coverage intensity.

- **Volatility and uncertainty are present.** Year-to-year fluctuations are visible across all countries, as reflected in the uncertainty bands. This suggests that sentiment is responsive to specific events—such as policy announcements, adoption milestones, or regulatory developments—rather than following a perfectly monotonic path.

- **Overall conclusion.** The weight of evidence supports the interpretation that CBDC sentiment, as captured through Bloomberg news coverage, has trended negatively across the major study countries from 2020 to 2025.

3.2 REAL-WORLD REASONS FOR DECLINING CBDC SENTIMENT

The fall in CBDC sentiment seen in this study is not random. It reflects real-world problems that countries are facing while trying to implement digital currencies.

3.2.1 Low Adoption and Limited Usage

One major reason is low adoption and limited usage. Even though many countries launched or tested CBDCs, people are not using them much. This is because existing options like cash, mobile payments, and bank apps already work well. As a result, CBDCs do not provide enough extra benefit to attract users (IMF, 2025; BIS, 2023).

3.2.2 Privacy and Trust Concerns

Another important issue is privacy and trust concerns. CBDCs create a digital record of transactions, which makes people worry about government surveillance and misuse of data. Reports from the IMF and other institutions highlight that this "data tracking" feature can reduce public trust if not handled properly (IMF, 2024). In fact, many experts

argue that CBDCs could allow governments to monitor financial activity, which increases fear among users (Cato Institute, 2023).

3.2.3 Policy and Political Concerns

There are also policy and political concerns, especially in countries like the United States. Even without launching a CBDC, debates around privacy, financial control, and regulation have created negative media coverage. Some policymakers believe CBDCs could "upend the banking system" or reduce financial freedom (Blockworks, 2023).

3.2.4 Technological and Infrastructure Challenges

Another key factor is technological and infrastructure challenges. CBDCs require strong digital systems, internet access, and merchant acceptance. In many countries, these systems are still developing, which limits usability and slows adoption. Reports from BIS show that CBDCs involve operational and technological risks that must be carefully managed (BIS, 2023).

3.2.5 Impact on the Banking System

There are concerns about the impact on the banking system. CBDCs may reduce the role of commercial banks because people can hold money directly with the central bank. This could affect lending and financial stability, making policymakers more cautious (Dionysopoulos, 2024).

3.2.6 Changing Media Narratives

Finally, media narratives have changed over time. In the beginning, CBDCs were described as innovative and beneficial. But as real-world challenges appeared, news coverage became more critical, focusing on risks, failures, and slow progress. Since this study uses Bloomberg news data, this shift in media tone directly leads to lower sentiment scores.

Overall, the decline in CBDC sentiment is mainly due to low adoption, privacy concerns,

policy debates, technological limitations, and changing media narratives. This shows that CBDCs have moved from a stage of hype to a stage of real-world evaluation, where problems are more visible.

3.3 FUTURE SCOPE

This project can be improved in several ways.

First, the dataset can be expanded by adding more Bloomberg articles and more time periods. This would make the results more reliable. More countries can also be included for comparison.

Second, the model can be improved by using stronger BERT-based methods. This may produce more accurate sentiment predictions for financial news.

Third, the study can be extended to examine different topics within CBDC news, such as privacy, government control, payment speed, and financial inclusion. This would give a deeper understanding of the news.

Fourth, more manual labeling can be done to improve the quality of the dataset. If more people review the sentences, the sentiment scores will be more trustworthy.

Fifth, the project can be linked with important CBDC events, such as policy announcements, pilot programs, or new rules. This can help explain why sentiment changes over time.

Finally, this work can be turned into a tool that updates Bloomberg-based CBDC sentiment results automatically and creates new charts. This would make it more useful for future research.

3.4 CONCLUSION

This study developed a structured and reliable framework for analyzing sentiment in CBDC-related financial news using sentence-level annotation and machine learning techniques. By combining human and model-based annotations, the research ensured high-quality sentiment scoring and produced meaningful country-level insights across the United States, Nigeria, and Jamaica.

The results clearly show that CBDC sentiment has declined over the period 2020–2025 in all three countries, with the sharpest drops observed in the United States and Nigeria. This confirms that public and media perception of CBDCs has shifted over time. Initially, CBDCs were seen as innovative tools with the potential to improve financial inclusion and payment systems. However, as real-world implementation progressed, sentiment became more negative.

This decline is supported by real-world evidence. Studies show that CBDC adoption remains low compared to existing payment systems, and many users prefer familiar options such as cash and digital banking. Additionally, concerns related to privacy, trust, and lack of awareness have created barriers to adoption, further reducing positive sentiment. Policy debates, especially in advanced economies, have also shifted attention toward risks such as financial surveillance and banking system disruption. Moreover, CBDCs introduce potential financial stability risks, including the possibility of users moving funds away from commercial banks, which adds to negative discourse.

Another key insight is the role of media framing. Since this study relies on Bloomberg news data, the sentiment scores reflect how CBDCs are portrayed in financial journalism. Over time, media coverage has shifted from focusing on innovation and benefits to highlighting challenges such as low adoption, technological limitations, and regulatory concerns. This change in narrative plays a significant role in the observed decline in sentiment.

Overall, the findings suggest that CBDCs have moved from a stage of early optimism to practical evaluation. The declining sentiment does not necessarily indicate failure, but rather a natural progression where initial expectations are tested against real-world performance. This highlights the importance of addressing adoption barriers, improving public trust, and clearly communicating the value of CBDCs.

In conclusion, this study not only provides a robust methodology for analyzing financial sentiment but also offers important insights into the evolving perception of digital currencies. Future research can build on this work by incorporating multiple news sources, expanding country coverage, and exploring the impact of specific policy events on sentiment dynamics.

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